

Room 315 Rail-Cell VLA Research Progress

Current Progress

- I have defined the general research direction for applying **Vision-Language-Action (VLA)** to the Room 315 rail-cell simulation.
- The current focus is on a **rail-only VLA stage**, before extending the work later to the four robot stations.
- I studied where VLA is actually useful, instead of using it only as an additional component.
- The main research motivation is that the rail system has **sparse binary sensors**, which do not provide continuous shuttle position.
- Therefore, overhead vision can help in cases where the state alone is ambiguous, such as obstacles, visual occupancy, or shuttle position between sensors.

Simulation and Visual Setup

- I planned the use of two overhead VLA cameras:
 - one camera above the right rail,
 - one camera above the left rail.
- I explored the use of visual elements in Gazebo, including slot markers, and visual obstacle markers.
- The final direction is to keep only meaningful visual cues and avoid artificial shortcuts that make the task unrealistically easy.
- The model should learn from the shuttle, rail geometry, slot fiducials, and visual-only obstacles, not from debug colors.

Proposed VLA Scenarios

- **Visible obstacle stop:** move the shuttle toward a target station, but stop before any obstacle that is visible in the overhead camera view. The obstacle is not provided in the structured state, so the model must rely on vision.
- **Shortest safe route:** choose the shortest available route to the target station while avoiding a visible obstacle. This scenario requires both route selection and visual hazard awareness.
- **Visual slot occupancy:** park the shuttle at a visually empty slot. The model should infer whether a slot is empty or occupied from the overhead image, for example by checking whether the slot fiducial is visible or covered.
- **Payload-aware routing:** route the shuttle based on whether it is empty or carrying a visible object. For example, move the empty shuttle to one station and the loaded shuttle to another station. The payload state should not be given as a structured input; it should be inferred from the camera image.
- **Sensor dropout:** continue routing safely even when one of the sparse position sensors is unavailable. In this case, the model should use visual information to compensate for missing or unreliable sensor feedback.

Methodology Defined

- The model input should include:
 - language goal,
 - overhead images,
 - binary sensor bits,
 - switch states,
 - stopper states,
 - last command type,
 - shuttle command state.
- Privileged simulation information, such as exact Gazebo pose, obstacle coordinates, true segment, or exact distances, should be used only for evaluation and expert data generation.

- The action formulation will be based on **event-level symbolic actions**, not repeated frame-by-frame commands.
- Each training row should represent:

$$observation_{beforedecision} \rightarrow nextsymbolicaction$$

- Human-readable actions will be used for inspection, while numeric **action_vector** representations will be used for training.

Evaluation Plan and Next Steps

- Compare a **state-only** model with a **VLA** model.
- The state-only model receives language and sparse rail state only.
- The VLA model receives language, sparse rail state, and overhead images.
- The expected result is that VLA should outperform state-only in visual obstacle, occupancy, and ambiguous-position scenarios.
- Next steps:
 - finalize the scenario design,
 - generate expert demonstrations,
 - record images, state, and event-level actions,
 - build a balanced dataset,
 - train and compare state-only and VLA baselines,
 - prepare the work for future LeRobot or SmolVLA integration.